

Indraprastha Institute of Information Technology, Delhi

IIIT-Delhi - Department of Computer Science & Engineering

Discrete Mathematics - Assignments

Course Resource Package

CSE111 - Discrete Math Assignment 1:

Q1. Prove by induction that $1^2 + 2^2 + \dots + n^2 = n(n+1)(2n+1)/6$.

Q2. Solve the recurrence relation: $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0=1$, $a_1=2$.

Q3. How many edges does a tree with n vertices have? Prove it.

Q4. Find chromatic number of complete bipartite graph $K_{(m,n)}$.